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Efficient crowd anomaly detection using C3D-LSTM networks with enhanced attention mechanisms

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ABSTRACT

The rising deployment of surveillance systems in urban environments necessitates efficient automated anomaly detection methods. While showing promise, current deep learning approaches struggle with computational complexity and real-time performance in processing spatiotemporal information. This paper presents a hybrid framework integrating Convolutional 3D Networks (C3D), Long Short-Term Memory (LSTM) networks, and attention mechanisms for anomaly detection. Through a systematic evaluation of four attention mechanisms—self-attention, multi-head attention, Bahdanau attention, and Luong attention—we demonstrate their operational differences and their differential impact on feature extraction and classification performance across three diverse benchmark datasets. Our multi-head attention variant achieves state-of-the-art results with 99.40 % accuracy and 99.96 % Area Under the Curve (AUC) on Violent Flows, while maintaining robust performance across varying dataset complexities, achieving 91.87 % accuracy on the ShanghaiTech Campus and 79.7 % accuracy on the UCF-Crime dataset. Comprehensive cross-dataset evaluation demonstrates consistent improvements of 2.4 %–3.5 % over baseline approaches, with all attention mechanisms outperforming traditional spatiotemporal models. The proposed architecture effectively balances computational requirements with detection performance, maintaining real-time processing capabilities suitable for operational deployment. This framework advances the technical capabilities of anomaly detection systems while providing a validated foundation for practical deployment in diverse surveillance environments, from controlled scenarios to challenging real-world conditions.

1. Introduction

The proliferation of surveillance systems in modern urban environments has generated an unprecedented volume of video data, presenting both opportunities and challenges for public safety monitoring. Developing autonomous systems capable of analysing continuous video streams to detect potential security threats to individuals, property, and public spaces has emerged as a critical challenge in computer vision research [1]. Traditional surveillance methods, which rely on human operators, face significant limitations due to the overwhelming volume of information, resulting in diminished vigilance and potential oversight of violent incidents. Further complexity arises from the dynamic nature of human interactions in urban settings, where suspicious or violent activities often involve intricate interactions between multiple individuals and their environment, requiring sophisticated analysis of spatial and temporal patterns [2].

Contemporary research has demonstrated that contextual information plays a crucial role in understanding these complex scenarios, particularly when conventional motion-based analysis fails to capture the full scope of developing situations. This is especially relevant for violent activities occurring in isolated or poorly illuminated environments where traditional surveillance proves less effective. Criminal behaviour in such contexts typically follows predictable patterns, with perpetrators conducting reconnaissance before executing their acts. This behavioural signature enables the development of preventive response systems based on automated detection, provided they can effectively interpret pre-criminal indicators.

Despite significant advances in computer vision, human action recognition presents persistent challenges in real-world scenarios characterised by complex backgrounds, occlusions, and variable lighting conditions. The potential applications extend beyond security to include industrial monitoring, cloud-based surveillance, identity verification,

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and virtual reality [3,4]. However, existing systems face three fundamental limitations:

1. Feature Representation Challenges: Traditional approaches rely on handcrafted features to convert video signals into three-dimensional representations of dynamic motion. While necessary for action description, these methods often fail to capture the full complexity and variability of human violent behaviour, particularly in crowded environments.
2. Contextual Understanding Deficiencies: Traditional action identification methods, predominantly based on database matching, struggle to adapt to the substantial real-world variability in movement styles, environmental conditions, and social contexts that characterise violent incidents.
3. Temporal Integration Limitations: Despite the evolution from 2D to 3D feature extraction techniques, existing approaches have not fully addressed the challenges of consistently representing temporal data and integrating information across varying time scales [5].

These limitations are particularly evident in violence detection, where the consequences of false negatives (missed detections) and false positives (false alarms) are incredibly high. The demand for systems that can operate reliably in diverse environmental conditions while maintaining real-time performance has driven recent innovations in deep learning approaches for video analysis.

This paper proposes a novel hybrid framework that addresses these challenges by integrating Convolutional 3D Networks (C3D), Long Short-Term Memory (LSTM) networks, and advanced attention mechanisms. Our approach makes several key contributions:

- Hierarchical Feature Integration: We introduce a comprehensive architecture that effectively bridges spatial and temporal feature extraction through a multi-stage pipeline, enabling more nuanced anomaly detection in complex scenarios.
- Attention Mechanism Analysis: We provide an in-depth comparative evaluation of four distinct attention mechanisms—self-attention, multi-head attention, Bahdanau attention, and Luong attention—quantifying their impact on detection performance and computational requirements.
- Performance Advancement: State-of-the-art results on standard benchmark datasets demonstrating significant improvements in both accuracy and computational efficiency
- Implementation Framework: A practical framework for early warning system implementation in real-world surveillance applications

The remainder of this paper is organised as follows: Section 2 reviews related work in video-based anomaly detection and attention mechanisms. Section 3 presents our proposed methodology and architectural details. Section 4 describes our experimental setup and the results obtained. Finally, Section 5 discusses our findings and their implications for future research.

2. Related work

Deep Learning (DL) approaches have emerged as the predominant methodology for extracting high-level discriminative features and developing comprehensive systems for video-based action and behaviour detection [6]. This section reviews key developments in deep learning architectures for human activity recognition, with particular emphasis on approaches relevant to anomaly detection.

2.1. Convolutional Neural Networks for spatial feature extraction

Traditional Convolutional Neural Networks (CNNs) have demonstrated considerable success in extracting spatial features from individual video frames. Landmark architectures, including AlexNet [7], ResNet [8], and VGG [9], have established a solid foundation for visual feature extraction. However, these standard CNN models primarily capture spatial properties from single frames, lacking the temporal modelling capability essential for Abnormal Human Activity Recognition (AbHAR) [10]. This limitation is particularly significant for anomaly detection, where the temporal evolution of motion patterns provides critical discriminative information.

2.2. 3D CNNs for spatiotemporal modelling

To address the temporal limitation of 2D CNNs, researchers explored 3D CNNs that capture spatial and temporal features jointly. Song et al. [11] proposed adapting the C3D model developed by Tran et al. [12], comprising eight 3D convolutional layers and five 3D pooling layers, with a novel frame sampling technique based on grey centroid analysis to reduce redundancy. This approach was validated across multiple benchmarks, including the Violence Flow, Hockey, and Movies datasets. Similarly, Ullah et al. [13] introduced a three-stage methodology combining person detection algorithms with 3D CNN for spatiotemporal feature extraction, followed by softmax classification. While outperforming classical approaches, these pure CNN-based methods still presented opportunities for improvement through integration with complementary architectures.

2.3. Hybrid CNN–RNN architectures

Recognising the complementary strengths of different neural network paradigms, researchers have explored hybrid architectures that combine CNNs with recurrent models. A particularly effective approach employs 2D time-distributed CNNs for initial feature extraction, followed by Recurrent Neural Networks (RNNs) for temporal refinement [14]. Simonyan et al. utilised pre-trained VGG19 [9] as a feature extractor, feeding the extracted representations into Long Short-Term Memory (LSTM) networks and time-distributed fully connected layers for final classification. This integrated approach achieved competitive results across standard anomaly detection benchmarks.

Feature extraction coupled with LSTM processing has become a foundational strategy for many CNN-RNN hybrid systems [15]. Implementation variations include processing entire frames through CNNs [16], dividing frames into patches to preserve discriminative features despite scale and position variations [17], and employing multi-scale convolutional feature extraction to accommodate videos of varying dimensions. Ditsanthia et al. [17] specifically demonstrated the effectiveness of feeding LSTMs with multi-scale extracted features to handle the inherent variability in surveillance video content.

Despite advances in hybrid architectures, challenges persist in modelling human movements that depend on specific motion characteristics, stylistic elements, and environmental factors. Conventional approaches struggle particularly with continuous activities and crowded scenes [18]. LSTMs and Gated Recurrent Units (GRUs) have shown promise in addressing sequence learning challenges by maintaining temporal context, but further refinement is needed to ensure coherent transitions between frames.

2.4. LSTM variants and applications in security contexts

LSTM units, introduced by Hochreiter and Schmidhuber [19], have been successfully integrated with RNNs across applications such as video captioning [20], speech recognition [21], and handwriting recognition. However, limitations in standard implementations—such as insufficient parameter sharing and restricted local connectivity—prompted the development of new LSTM topologies.

Alex Graves [22] demonstrated how bidirectional LSTM networks could effectively categorise phonemes in real-time speech detection, while Hasim Sak [23] proposed LSTM variants for large-scale acoustic modelling. These studies provided empirical evidence for LSTM's effectiveness in representing long-term temporal dependencies across diverse tasks.

Building on these foundations, researchers applied LSTM-based frameworks to anomaly detection in surveillance and security contexts. Ansari et al. [24] developed a system that combines the Inception and LSTM frameworks to detect shoplifting in retail environments, achieving 91.8 % accuracy on a custom dataset. In follow-up work, Ansari et al. [25] integrated gradient and optical flow data to identify salient motion characteristics for precise shoplifting detection. Similarly, Dwivedi et al. [26] proposed a novel hybrid approach that combines pre-trained CNNs with LSTMs for detecting suspicious activity across eleven benchmark datasets.

2.5. *Attention mechanisms in activity recognition*

Attention-based systems represent a significant advancement for AbHAR applications. By focusing on distinctive cues in temporal streams, these systems can more effectively extract relevant spatiotemporal properties from video sequences. Deep Convolutional Neural Networks (DCNNs) enhanced with attention blocks improve feature learning, while Bi-directional LSTMs (Bi-LSTMs) with attention weights enable more focused analysis of motion patterns across frame sequences [27,28].

Feedforward neural networks incorporating attention mechanisms have achieved remarkable object classification accuracy, in some cases surpassing human performance. Baccouche et al. [29] and Latah [30] presented multiple 3D deep learning models and 3D CNNs specifically tailored for Human Activity Recognition, demonstrating strong performance on the KTH dataset [31]. Jaouedi et al. [32] further validated the effectiveness of deep learning approaches for video classification tasks.

2.6. *Weakly supervised and crowd-based anomaly detection*

Recent advances in weakly supervised and attention-enhanced approaches have further expanded the capabilities of anomaly detection systems. Lv et al. [33] introduced Unbiased Multiple Instance Learning (UMIL) for weakly supervised video anomaly detection, addressing bias where normal frames receive disproportionate attention. Their objective instance selection mechanism ensures balanced treatment of normal and anomalous frames during training. Zhou et al. [34] developed a complementary approach using Dual Memory Units (DMUs) with uncertainty regulation to store and process normal and anomalous patterns separately, improving robustness when only video-level labels are available.

Crowd analysis has also gained attention and research on anomaly detection in crowd scenes via cross trajectories, demonstrating the value of analysing trajectory intersections to detect congestion or unusual group formations. Mahmoud et al. [35] developed a two-stage framework for crowd violence detection, combining Global Motion Flow (GMFlow) with CBAM-enhanced ResNet3D. Their system achieved 94.2 % accuracy, highlighting the benefits of combining motion analysis with attention-enhanced spatiotemporal learning.

Recent advances (2024–2025) in weakly supervised video anomaly detection (WSVAD) highlight the field's movement toward prompt-based learning, multimodal integration, and advanced representation schemes. Notable examples include TPWNG [36], LEC-VAD [37], Hyperbolic WSVAD [38], and ProDisc-VAD [39]. A detailed comparative analysis of these methods, including their benchmark performance, is presented in Sec. 4.5.

2.7. *Summary and positioning of our work*

These recent developments highlight the importance of attention mechanisms and hybrid architectures in video anomaly detection. The present research builds upon these foundations by systematically evaluating four distinct attention mechanisms within a unified C3D–LSTM framework. Unlike prior studies, which often focus on stacking components, our analysis dissects the mechanism-level (operational) principles of each attention variant and quantifies their impact on anomaly detection performance. This provides comprehensive insights into their relative effectiveness for surveillance applications, advancing the state of the art in automated surveillance for public safety.

3. Proposed method

Modern video surveillance systems require robust and efficient anomaly detection capabilities that can process complex spatiotemporal patterns while maintaining real-time performance. Our proposed architecture addresses this challenge by integrating deep learning components specifically designed for anomaly detection. The system uniquely combines multiscale temporal processing with adaptive attention mechanisms to capture both immediate violent actions and their contextual evolution.

3.1. *Architecture overview*

The proposed architecture implements a hierarchical feature processing pipeline that analyses video sequences at multiple temporal scales and levels of abstraction. As illustrated in Fig. 1, the system processes surveillance video through a structured pipeline with four main components: Input Processing, Spatiotemporal Feature Extraction, Temporal Analysis, and Attention Enhancement, culminating in final Classification.

The Input Processing stage, shown in the upper-left component of Fig. 1, handles continuous video streams by segmenting them into sequences of 16 frames, each standardised to $64 \times 64 \times 3$ dimensions. This standardisation ensures consistent processing while preserving essential motion and appearance information. The frame sequence length of 16 was empirically determined to provide sufficient temporal context for anomaly detection while maintaining computational efficiency.

Following input processing, the raw frame sequences are passed to the Spatiotemporal Feature Extraction stage, displayed in the lower-left component of Fig. 1. This stage employs a Conv3D Network comprising 8 Conv3D layers and 5 MaxPool3D layers to capture localised motion patterns. Unlike traditional 2D CNNs that process frames independently, the Conv3D network analyses temporal neighbourhoods of frames simultaneously, enabling the detection of motion patterns characteristic of violent behaviour.

The Temporal Analysis component, shown in the centre of Fig. 1, addresses the limitation of Conv3D in capturing longer-term temporal dependencies by incorporating an LSTM Network. This network processes the spatiotemporal features sequentially, leveraging memory states to track long-term patterns. While Conv3D excels at detecting immediate violent actions through local spatiotemporal patterns, the LSTM network maintains memory of past events and tracks the development of potentially violent situations. This dual-temporal processing approach enables the system to analyse both immediate actions and their broader temporal context.

The Attention Enhancement stage, depicted in the right-centre of Fig. 1, implements attention mechanisms that dynamically weigh feature importance based on spatial and temporal context. This component performs feature refinement through dynamic weighting and context focus, helping the system concentrate on relevant participants and actions while filtering out irrelevant background motion. We implement and compare four different attention mechanisms—self-attention, multi-head attention, Bahdanau attention, and Luong attention—to determine the most effective approach for anomaly detection. Finally, as shown in the upper-right component of Fig. 1, the Classification stage produces anomaly detection output with binary classification results and associated confidence scores, based on the refined features generated by the attention mechanism.

A key innovation in our architecture lies in how these components

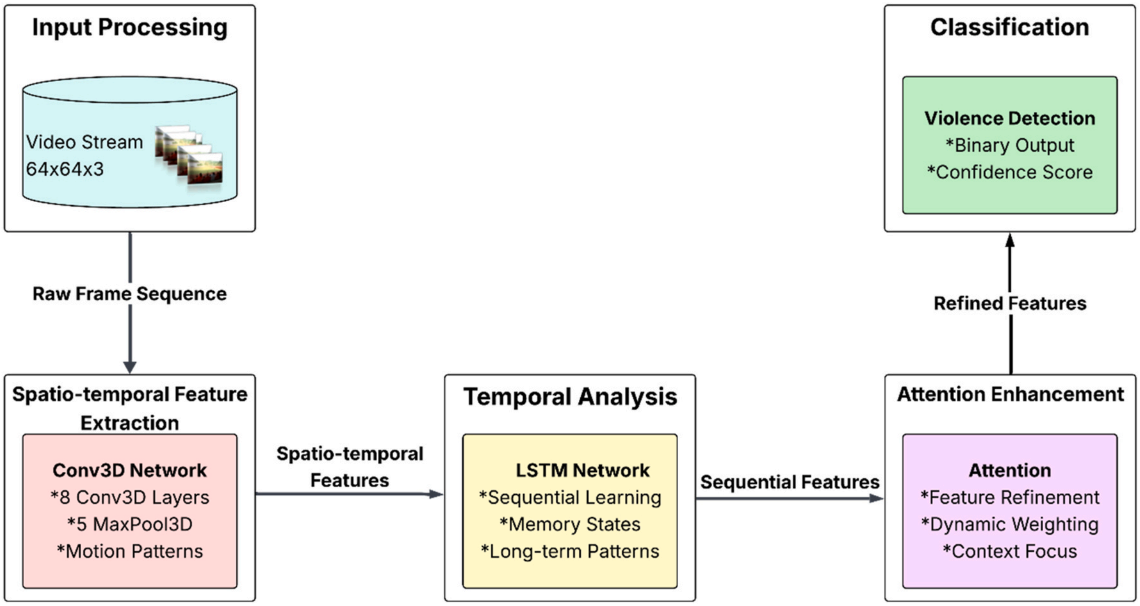


Fig. 1. The proposed architecture for anomaly detection.

are integrated. Rather than simply concatenating features from different processing stages, we implement a hierarchical feature fusion strategy. C3D features provide the foundation for understanding motion patterns, which are then refined by LSTM states with temporal context and finally weighted by the attention mechanism based on their relevance to anomaly detection. This progressive refinement ensures that spatial and temporal information are effectively utilised in the final classification decision.

The architecture also incorporates several violence-specific optimisations. The C3D kernel sizes are modified to better capture the motion patterns characteristic of violent actions. The LSTM gates are tuned to forget meaningless temporal data while preserving memory of pertinent behavioural patterns. The attention mechanisms are weighted toward traits usually indicating aggressive behaviour. These improvements help the system detect violent events while maintaining strong performance in demanding real-world conditions.

3.2. Technical components

The effectiveness of our proposed architecture relies on three main technical components: 3D Convolutional Neural Networks for extracting spatial-temporal features, Long Short-Term Memory networks for modelling long-term temporal dependencies, and attention mechanisms for focused feature selection. Each component addresses specific challenges in anomaly detection, and their integration enables comprehensive analysis of violent behaviour in video sequences.

3.2.1. Convolutional neural network

3D CNNs extend traditional 2D convolution operations by incorporating temporal information through 3D convolution and 3D pooling operations. This architecture enables the network to capture spatial patterns within individual frames and temporal patterns across consecutive frames, which is crucial for anomaly detection in video sequences.

The 3D convolution operation processes a cube of data formed by stacking consecutive video frames. When this cube is convolved with 3D kernels, the resulting feature maps contain information about both spatial patterns and motion dynamics. This is formalised in Equation (1):

$$v_{ij}^{xyz} = tanh\left(b_{ij} + \sum_{m}\sum_{p=0}^{P_i-1}\sum_{q=0}^{Q_i-1}\sum_{r=0}^{R_i-1} w_{ijm}^{pqr} v_{(i-1)m}^{(x+p)(y+q)(z+r)}\right) \tag{1}$$

Where v_{ij}^{xyz} represents the output value at position (x,y,z) in the j-th feature map of the i-th layer, b_{ij} denotes the feature map's bias, P_i and Q_i denote the 3D kernel's spatial dimensions (height and width), R_i denotes the kernel's temporal dimension, and w_{ijm}^{pqr} denotes the (p,q,r)-th weight connected to the m-th feature map of the preceding layer. This formulation, as demonstrated by Ji et al. [40], enables the network to model spatiotemporal patterns characteristic of violent actions effectively. The 3D pooling operation extends the concept of spatial pooling to include the temporal dimension, thereby reducing computational complexity while preserving essential motion information. This hierarchical structure of 3D convolution and pooling operations allows the network to progressively build more abstract representations of violent actions from raw video frames.

3.2.2. Long Short-Term Memory (LSTM)

While 3D CNNs excel at capturing local spatiotemporal patterns within short sequences of frames, they are limited in their ability to model long-term temporal dependencies that often characterise the progression of violent incidents. Violence in crowds frequently evolves through distinct phases, from initial tension to escalation and eventual outbreak, spanning longer temporal intervals than what 3D CNN kernels can effectively capture. LSTM networks address this limitation by maintaining an internal memory state that can persist over extended sequences, making them complementary to the local feature extraction capabilities of 3D CNNs.

The LSTM [41] architecture, visualised in Fig. 2, comprises three gates (input, forget, and output) that regulate information flow through the network. Each gate learns to selectively remember or forget information based on its relevance to anomaly detection. The following Equations (2)–(7) define the operations of these gates:

Forget Gate: Controls what information to discard from the cell state, as shown in Equation (2):

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \tag{2}$$

Input Gate: Determines what new information to store in the cell state, defined in Equation (3):

$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i) \tag{3}$$

Output Gate: Controls what parts of the cell state are output, as formulated in Equation (4):

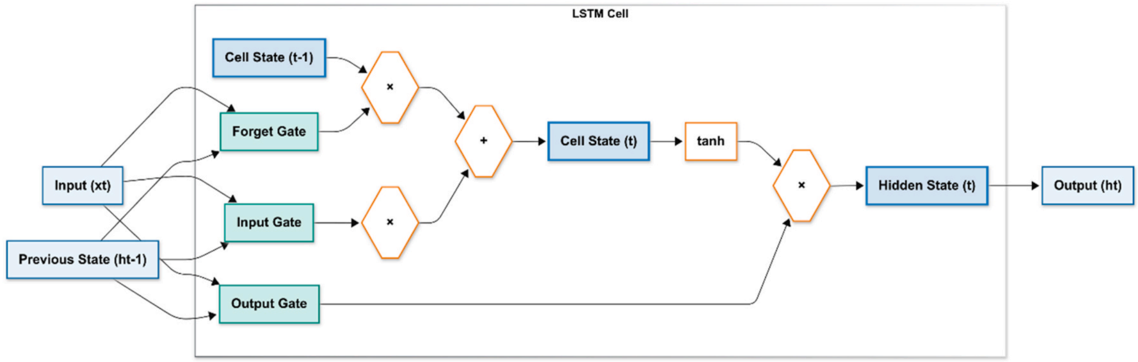


Fig. 2. LSTM architecture showing interactions between memory (cell state) and gates.

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o) \tag{4}$$

Cell Input: Generates candidate values for updating the cell state according to Equation (5):

$$\widetilde{c}_t = \sigma_h(W_c x_t + U_c h_{t-1} + b_c) \tag{5}$$

Cell State Update: Updates the long-term memory state as shown in Equation (6):

$$c_t = f_t \circ c_{t-1} + i_t \circ \widetilde{c}_t \tag{6}$$

Hidden State Output: Produces the final output as defined in Equation (7):

$$h_t = o_t \circ \sigma_h(c_t) \tag{7}$$

Where: x_t is the input vector at time t, h_t is the hidden state vector, c_t is the cell state vector, $\widetilde{c}_t$ is the candidate cell input, W and U are weight matrices learned during training, b represents bias terms, σ_g is the sigmoid activation function, σ_h is the hyperbolic tangent (tanh) activation and $\circ$ denotes element-wise multiplication,

As illustrated in Fig. 2, the cell state acts as a conveyor belt, carrying information, while gates control the flow of information through multiplicative interactions ($\times$). The forget gate filters old information, the input gate adds new information, and the output gate controls what information becomes the output. Mathematical operations (+, $\times$, tanh) show exactly how information is transformed at each step.

This gating mechanism enables LSTMs to learn which temporal features are crucial for anomaly detection and should be preserved across long sequences, and which can be discarded. For example, while analysing a crowd scene, the LSTM can maintain memory of gradual behavioural changes that might indicate impending violence, such as increasing agitation or formation of hostile groups, while the 3D CNN focuses on immediate physical actions. This dual temporal processing at different scales—local patterns through 3D CNN and long-term dependencies through LSTM—gives our system a more comprehensive understanding of violent behaviour development.

3.2.3. Attention mechanisms

Attention mechanisms enhance our model's ability to focus on the most relevant parts of input sequences for anomaly detection. In surveillance videos, violent events often involve complex interactions between multiple actors and may occur in small regions of the frame. Attention allows the model to dynamically weight different spatial and temporal features based on their relevance to anomaly detection, similar to how human observers focus on suspicious behaviours or sudden movements in a crowd.

The attention mechanism works by computing importance weights for different parts of the input sequence. These weights determine the extent to which each input element contributes to the output. For example, when analysing a sequence of frames, the attention mechanism might assign higher weights to frames showing sudden movements or aggressive gestures while giving lower weights to frames showing normal crowd behaviour.

In deep learning, attention mechanisms were popularised by the Transformer architecture, which revolutionised natural language processing and has since been adapted for computer vision tasks. Transformers replace traditional recurrent architectures with self-attention layers that can model relationships between all elements in a sequence, regardless of their relative distance from one another. This property is particularly valuable for anomaly detection, where intervening frames of normal activity may separate relevant events.

Fig. 3 provides a comparative visualisation of the four attention mechanisms implemented in our framework, showing their distinctive processing patterns.

As depicted in Fig. 3, each mechanism processes the video sequence differently: (1) Self-Attention transforms the input into Query-Key-Value representations and computes weighted relationships, (2) Multi-Head Attention performs parallel attention operations for comprehensive feature analysis, (3) Bahdanau Attention uses additive scoring for temporal alignment, and (4) Luong Attention employs multiplicative scoring with a flexible focus range. All mechanisms produce weighted features that highlight relevant aspects of violent behaviour.

To understand how each mechanism contributes to our anomaly detection framework, we examine their operational principles and specific advantages in detail:

3.2.3.1. *Self-attention.*

Self-attention [42] is the model to assess the relationships between different elements within an input sequence. It computes attention weights using three matrices derived from the input: Queries (Q), Keys (K), and Values (V). The attention output is calculated as Equation (8):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{8}$$

The matrices, K, and V are derived from the input sequence through linear transformations, where d_k represents the dimensionality of the keys. The dot product QK^T is used to determine the similarity between the query and key vectors and is scaled by $\sqrt{d_k}$ to prevent large values that could distort the softmax function. The softmax function normalises the attention weights to sum to 1, ensuring that they represent relative importance. The resulting attention weights are then multiplied by V to create a weighted representation.

In the context of anomaly detection, self-attention helps the model identify temporal relationships between actions across frames, such as the buildup of aggressive behaviour or the coordination of multiple participants in a violent incident.

3.2.3.2. *Multi-head attention.*

Multi-head attention, first proposed in the seminal work on transformer architecture by Vaswani et al. [42], is a

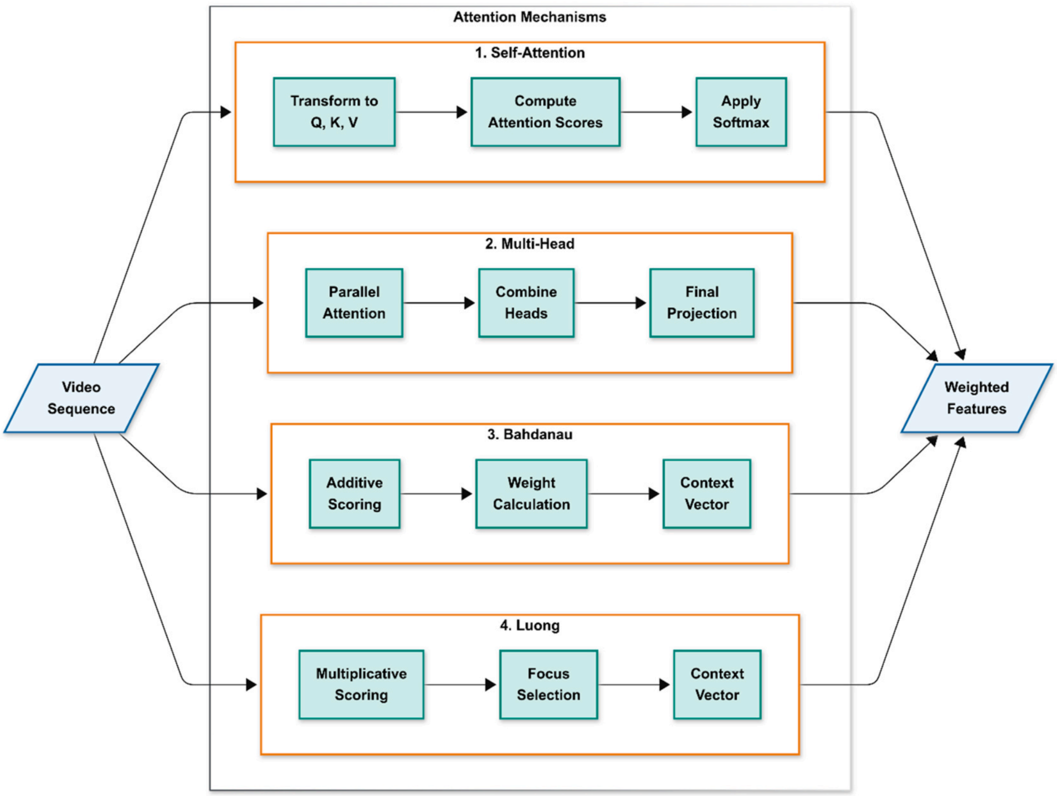


Fig. 3. Comparative view of the four attention mechanisms in our anomaly detection framework.

fundamental mechanism that enables models to capture diverse relationships between sequence elements across different representation subspaces. It extends the basic scaled dot-product attention by applying multiple parallel attention mechanisms, known as "heads," to the same input sequence. Each attention head computes attention independently using its own unique set of learned linear projections, allowing it to focus on different aspects of the relationships between sequence elements. For example, one head might attend to syntactic relationships while another focuses on semantic dependencies. The outputs from all attention heads are then concatenated and passed through a final linear projection to produce the consolidated representation that combines these multiple perspectives, as defined in Equation (9):

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O \tag{9}$$

Where each head is computed according to Equation (10):

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{10}$$

W^O is the projection matrix applied after concatenation, and $W_i^Q, W_i^K,$ $and\ W_i^V$ are the learned projection matrices specific to each head. The mechanism enables each head to focus on distinct parts of the sequence, resulting in a more comprehensive representation. In anomaly detection, different attention heads can specialise in different aspects of the scene - one head might focus on rapid movements, another on group formations, and another on individual actions, as illustrated in the second panel of Fig. 3.

3.2.3.3. Bahdanau Attention or additive attention. Bahdanau attention [43], also known as additive attention, is frequently employed in sequence-to-sequence models. It processes the relationship between the current frame features and the previous temporal context, and the attention scores are computed as shown in Equation (11):

$$e_{ti} = score(s_{t-1}, h_i) = v^T\ tanh\ (W_s s_{t-1} + W_h h_i) \tag{11}$$

Where W_s, W_h, and v are learned parameters, attention weights are computed by merging the decoder's hidden state s_{t-1} with each encoder's hidden state h_i. By applying a softmax function to normalise the scores, the attention weights α_{ti} are acquired as defined in Equation (12):

$$\alpha_{ti} = \frac{exp\ \ exp\ (e_{ti})}{\sum_{j=1}^{n} exp\ (e_{tj})} \tag{12}$$

At each time step in the decoder, the context vector c_t is calculated using Equation (13):

$$c_t = \sum_{i=1}^{n} \alpha_{ti} h_i \tag{13}$$

This mechanism, depicted in the third panel of Fig. 3 and formalised in Equations (11)–(13), is particularly effective at identifying key moments in the temporal progression of violent events, as it can learn to focus on frames where significant changes in behaviour occur.

3.2.3.4. Luong Attention or Multiplicative Attention. Luong Attention or (Multiplicative Attention) [44] differs slightly from Bahdanau attention in calculating attention weights by a multiplicative score function between encoder and decoder hidden states, as shown in Equation (14):

$$e_{ti} = h_t^T W h_i \tag{14}$$

Where the learned weight matrix is represented by W. The attention weights are produced by normalising the scores e_{ti} using a SoftMax function, as defined in Equation (15)

$$\alpha_{ti} = \frac{exp\ \ exp\ (e_{ti})}{\sum_{j=1}^{n} exp\ (e_{tj})} \tag{15}$$

The procedure for forming the context vector c_t is then identical to that of Bahdanau attention, as specified in Equation (16):

$$c_t = \sum_{i=1}^{n} \alpha_{ti} h_i \tag{16}$$

As illustrated in the fourth panel of Fig. 3, the Luong attention mechanism consists of three main processing steps: Multiplicative Scoring, Focus Selection, and Context Vector generation. The process begins with computing attention scores using the multiplicative approach defined in Equation (14). The Focus Selection stage then determines which parts of the input sequence should receive greater attention, functioning similarly to a spotlight that highlights the most relevant regions of the video sequence. Finally, the Context Vector stage aggregates this weighted information into a comprehensive representation that emphasises the most relevant features for anomaly detection.

Luong et al. [44] proposed two variants of the mechanism: i) global attention, which considers all hidden states in the input sequence for the attention mechanism, and ii) local attention, which only considers a subset within a defined window. Either approach is valid; however, the local attention approach is particularly relevant for anomaly detection as it can focus on specific temporal regions where suspicious activity may be occurring, reducing computational overhead from processing irrelevant frames.

Each attention mechanism offers distinct advantages for anomaly detection:

- Self-attention excels at capturing complex spatial relationships
- Multi-head attention enables parallel processing of different behavioural aspects
- Bahdanau attention is effective at tracking temporal progression
- Luong attention provides computational efficiency for real-time applications

These four attention mechanisms, working in conjunction with the CNN and LSTM components presented in Fig. 1, create a comprehensive anomaly detection system capable of analysing both spatial and temporal aspects of crowd behaviour. The pathway from video sequence to weighted features, clearly illustrated in Fig. 3, demonstrates how each mechanism processes the input differently yet produces refined representations that emphasise the most relevant aspects for anomaly detection.

In summary, while all four mechanisms ultimately produce weighted feature representations, their underlying operational principles differ significantly. Self-attention models global dependencies across the entire sequence. Multi-head attention broadens this capability by distributing focus across multiple specialised subspaces. Bahdanau attention introduces additive alignment to emphasise gradual temporal transitions, and Luong attention employs multiplicative scoring for sharper, often more localised focus. These mechanistic distinctions directly influence how anomalies are detected in different contexts—for example, multi-head attention can simultaneously capture group dynamics and individual behaviours, whereas Bahdanau attention may be more effective in identifying transitional moments of escalating violence. By explicitly evaluating these variants within the same C3D–LSTM framework, our study goes beyond component stacking to investigate how different operational mechanisms shape anomaly detection performance. A detailed comparative evaluation of these four attention mechanisms, highlighting both their quantitative performance and qualitative operational distinctions, is presented in Sec. 4.

4. Results and discussion

This section presents a comprehensive evaluation of our proposed anomaly detection framework across three benchmark datasets. We first describe our experimental setup, including dataset preparation, implementation details, and training methodology. We then define the evaluation metrics used to assess model performance, followed by a detailed analysis of results and comparisons with state-of-the-art approaches. Finally, we discuss the limitations of our current approach and identify promising directions for future research. Our experiments demonstrate consistent improvements across all datasets while maintaining real-time processing capabilities.

4.1. Experimental setup

The effectiveness of deep learning models depends significantly on the quality of the experimental setup, including dataset curation, implementation architecture, and training parameters. In this section, we detail these components to ensure reproducibility and establish a solid foundation for evaluating our proposed attention-enhanced anomaly detection framework.

4.1.1. Dataset description

The Violent Flows Dataset [45] serves as our primary experimental resource for videos of real-life crowd violence. During preliminary analysis, we identified significant data redundancy issues, including duplicate videos that differ by only a few frames or contain nearly identical sequences. To reduce bias and enhance model efficiency, we conducted a comprehensive de-duplication process involving video hash comparisons and temporal alignment techniques. The resulting curated dataset consists of 224 unique videos, with clip lengths ranging from 1.04 to 6.52 s (mean length: 3.60 s). All video frames were pre-processed and resized to 64×64 pixels to maintain consistency across the dataset. We employed a stratified partitioning approach, allocating 70 % to training, 10 % to validation, and 20 % to testing, while maintaining class balance across all subsets.

UCF-Crime [46] An extensive dataset comprising 1900 unedited real-world surveillance videos totalling 128 h. Following conventional division: 1610 videos for training and 290 videos for testing. This dataset presents challenges with diverse crime types and varying video quality.

ShanghaiTech campus [47] Collected at ShanghaiTech University, encompassing 13 distinct scenes with diverse lighting conditions and camera angles, comprising 130 anomalous events. Following Zhong et al.'s restructuring, we used 238 training videos and 199 testing videos.

All the datasets present several challenging characteristics inherent to real-world surveillance footage, including variable lighting conditions, camera motion, diverse crowd densities, and different manifestations of violent behaviours. These attributes make it particularly suitable for evaluating anomaly detection systems intended for practical deployment, as models must develop robustness against these variations to achieve reliable performance in actual surveillance applications.

4.1.2. Implementation details

Our implementation integrates advanced neural network components for detecting violent behaviours in video sequences. The architecture processes video frames through a pipeline that combines convolutional, recurrent, and attention layers to capture both spatial and temporal information effectively.

Each frame sequence is treated as a 5D tensor with dimensions (batch_size, sequence_length, height, width, channels). In our configuration, this is (batch_size, 16, 64, 64, 3), where 16 represents the number of consecutive frames, 64×64 the spatial resolution, and 3 the RGB channels. This structure preserves both temporal order and spatial information.

The feature extraction process begins with Conv3D layers, which capture local motion patterns across adjacent frames. ZeroPadding3D is applied in the initial phase to preserve boundary information, ensuring edge features are not degraded during convolution. The Conv3D and MaxPool3D layers are configured as mentioned below in Table 1:

While Conv3D excels at learning localised spatiotemporal features, it

Table 1
Conv3D and MaxPool3D layer configurations used in the proposed architecture.

Layer	Kernel size	stride	padding	Number of filters
Conv3D-1	$3 \times 3 \times 3$	$1 \times 1 \times 1$	same	32
Maxpool3D-1	$2 \times 2 \times 2$	$2 \times 2 \times 2$	valid	
Conv3D-2	$3 \times 3 \times 3$	$1 \times 1 \times 1$	same	64
Maxpool3D-2	$1 \times 2 \times 2$	$1 \times 2 \times 2$	valid	

is limited in capturing long-range temporal dependencies. To address this, we integrate an LSTM layer after convolution, enabling the model to contextualise immediate actions within longer temporal sequences.

On top of the LSTM, we incorporate attention mechanisms (self-attention or multi-head attention), which prioritise the most critical temporal segments while down-weighting less relevant information. This three-tier refinement process (Conv3D → LSTM → Attention) ensures:

1. Conv3D extracts localised spatiotemporal patterns,
2. LSTM models long-term temporal dynamics, and
3. Attention highlights salient temporal dependencies.

Finally, dense layers with a sigmoid activation function perform binary classification of violent vs. non-violent behaviours.

4.1.3. Training parameters

The model training was conducted using the Adam optimiser (Adaptive Moment Estimation), an extension of stochastic gradient descent that adapts learning rates for each parameter by estimating first and second moments of the gradients [48]. This optimiser effectively handles sparse gradients and noisy data, making it particularly suitable for video analysis tasks. We configured Adam with a learning rate of 0.001. We set the batch size to 10, balancing memory constraints against training stability. The networks were trained for 50 epochs with early stopping to ensure convergence while avoiding overfitting.

4.1.4. Evaluation metrics

To quantitatively assess the performance of our anomaly detection framework, we employ multiple complementary metrics that capture different aspects of model performance. These metrics were carefully selected to provide a comprehensive evaluation of classification accuracy and discriminative capability, particularly in the context of surveillance applications where both false positives and false negatives carry significant consequences.

Accuracy provides an intuitive measure of the model's overall performance by calculating the ratio of correct predictions to the total number of predictions. For binary classification tasks like violent scene detection, accuracy is calculated as shown in Equation (17):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

Where:

- TP (True Positive): Correctly identified violent scenes
- TN (True Negative): Correctly identified non-violent scenes
- FP (False Positive): Non-violent scenes misclassified as violent
- FN (False Negative): Violent scenes misclassified as non-violent

While accuracy provides a straightforward assessment of model performance, it can be misleading for imbalanced datasets where one class (typically non-violent scenes) significantly outnumbers the other. To overcome this limitation, we also apply the Area Under the Curve (AUC-ROC) [49,50].

Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) curve quantifies a model's ability to discriminate between classes across all possible classification thresholds. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings, as defined in Equations (18) and (19):

$$TPR = \frac{TP}{TP + FN} \tag{18}$$

$$FPR = \frac{FP}{FP + TN} \tag{19}$$

The AUC-ROC value ranges from 0 to 1, where 1 represents perfect classification, 0.5 indicates random chance performance (equivalent to a random classifier), and values below 0.5 suggest worse-than-random performance. A higher AUC value indicates better model discrimination between violent and non-violent classes.

AUC-ROC measures how well a model can distinguish between classes by examining the relationship between True Positive Rate (sensitivity) and False Positive Rate across different classification thresholds. Since AUC-ROC calculates its results regardless of the thresholds used, it is most effective for use in anomaly detection, where the best decision may differ from one application situation to another.

These metrics provide a complementary view of model performance accuracy. Accuracy is an intuitive quantitative measure of performance, while AUC-ROC measures the model's ability to discriminate across classes, regardless of their distribution or any threshold set. This approach with two measures is consistent with recent studies [51] on crowd behaviour analysis and allows a good performance evaluation under different implementation conditions.

4.2. Performance analysis by dataset

Our experimental evaluation demonstrates the substantial performance benefits of integrating attention mechanisms with traditional spatiotemporal models for anomaly detection. In this section, we analyse the performance of different model configurations on the three datasets, compare our approach with state-of-the-art methods, and discuss both the strengths and limitations of our proposed framework.

4.2.1. Violent Flows Dataset results

Table 2 presents comprehensive performance metrics for each attention mechanism variant on the Violent Flows dataset, evaluated using a 5-fold cross-validation approach. The experimental results demonstrate the effectiveness of our proposed hybrid architecture across different attention mechanisms. All models were trained on the same training set (70 % of the dataset), validated using the same validation set (10 %) and evaluated on the test set (20 % of the dataset). We complemented accuracy metrics with AUC-ROC evaluation due to its robustness to class imbalance and threshold-independence, which is crucial for anomaly detection tasks where the distribution of violent and non-violent scenes may be uneven.

The Violent Flows dataset demonstrates exceptional performance improvements with attention integration. Multi-head attention achieves the highest metrics (99.40 % accuracy, 99.96 % AUC), representing a +2.9 % accuracy improvement over the baseline. Self-attention delivers the second-best performance (+1.81 % accuracy), while Bahdanau and Luong mechanisms provide consistent gains of +1.04 % and +1.49 %, respectively. The consistently low standard deviations (<0.5 %) across all models indicate robust generalisation capabilities, with multi-head attention showing the most stable performance.

Table 2
Performance Comparison on Violent Flows dataset.

Model Configuration	Accuracy (%)	AUC (%)
C3D + LSTM (baseline)	96.50 ± 0.45	99.00 ± 0.32
C3D + LSTM + Self-Attention	98.31 ± 0.38	99.79 ± 0.28
C3D + LSTM + Multi-head Attention	99.40 ± 0.35	99.96 ± 0.25
C3D + LSTM + Bahdanau Attention	97.54 ± 0.42	99.58 ± 0.30
C3D + LSTM + Luong Attention	97.99 ± 0.40	99.85 ± 0.29

4.2.2. ShanghaiTech Campus Dataset results

Table 3 presents performance metrics on the ShanghaiTech Campus dataset, which introduces significant complexity through varied lighting conditions, diverse camera angles, and subtle anomaly types across 13 distinct campus scenes.

ShanghaiTech introduces significant complexity through varied lighting conditions, diverse camera angles, and subtle anomaly types across 13 distinct campus scenes. Multi-head attention achieves the highest performance (91.87 % accuracy, 90.47 % AUC), representing a substantial +2.42 % accuracy and +5.34 % AUC improvement over baseline. Bahdanau attention demonstrates competitive accuracy performance (91.81 %, +2.36 % improvement) but lower discriminative capability (85.06 % AUC). Interestingly, self-attention shows minimal improvement (−0.04 % accuracy) and Luong attention delivers only modest gains (+0.53 % accuracy), with both mechanisms showing reduced AUC performance. This pattern suggests that complex multi-scene environments particularly benefit from multi-head attention's parallel processing capabilities for handling diverse visual contexts simultaneously.

4.2.3. UCF-Crime Dataset results

Table 4 presents the results on the UCF-Crime dataset, the most challenging benchmark with 1900 surveillance videos containing diverse crime types and real-world conditions.

The UCF-Crime dataset represents the most challenging evaluation scenario with diverse crime categories, varying video quality, and real-world surveillance conditions. Despite lower absolute performance metrics compared to controlled datasets, attention mechanisms demonstrate consistent and substantial improvements. Multi-head attention achieves the highest performance (79.7 % accuracy, 87.52 % AUC), delivering a significant +3.5 % accuracy improvement over the baseline. All attention variants provide meaningful enhancements: Bahdanau attention (+2.8 % accuracy), Luong attention (+2.4 % accuracy), and self-attention (+1.9 % accuracy). The consistent improvement pattern across all mechanisms validates the robustness of attention integration for real-world deployment scenarios, where diverse crime types and challenging recording conditions are prevalent.

4.3. Cross-dataset performance analysis

4.3.1. Attention mechanism effectiveness

Fig. 4 illustrates the comparative performance of different attention mechanisms across all three datasets. Accuracy improvement percentages across three benchmark datasets. Multi-head attention demonstrates the most consistent improvements (+2.4 % to +3.5 %), while all attention mechanisms show positive performance gains, validating the effectiveness of attention integration in spatiotemporal feature learning.

Performance Consistency: Multi-head attention consistently outperforms other variants across all datasets, with improvements of:

- Violent Flows: +2.90 % accuracy, +0.96 % AUC over baseline
- ShanghaiTech: +2.42 % accuracy, +5.34 % AUC over baseline
- UCF-Crime: +3.50 % accuracy, +3.37 % AUC over baseline

To further validate multi-head attention's superior discriminative capability, Fig. 5 presents the ROC curves for this optimal configuration across all three datasets.

Table 3
Performance comparison on Shanghai Tech campus dataset.

Model Configuration	Accuracy (%)	AUC (%)
C3D + LSTM (baseline)	89.45	85.13
C3D + LSTM + Self-Attention	89.41	84.43
C3D + LSTM + Multi-head Attention	91.87	90.47
C3D + LSTM + Bahdanau Attention	91.81	85.06
C3D + LSTM + Luong Attention	89.98	79.74

Table 4
Performance comparison on UCF-crime dataset.

Model Configuration	Accuracy (%)	AUC (%)
C3D + LSTM (baseline)	76.2	84.15
C3D + LSTM + Self-Attention	78.1	85.89
C3D + LSTM + Multi-head Attention	79.7	87.52
C3D + LSTM + Bahdanau Attention	79.0	86.44
C3D + LSTM + Luong Attention	78.6	86.02

The ROC analysis confirms multi-head attention's exceptional discriminative performance across varying dataset complexities. On the Violent Flows dataset, the ROC curve approaches the ideal top-left corner, achieving 99.96 % AUC with near-perfect true positive rates while maintaining minimal false positive rates across all decision thresholds.

For ShanghaiTech Campus, despite the increased complexity of subtle anomalies across 13 different scenes, multi-head attention maintains strong discriminative capability (90.47 % AUC). The ROC curve demonstrates consistent performance across various threshold settings, crucial for adaptable deployment in different campus security scenarios.

The UCF-Crime ROC curve illustrates the robustness of multi-head attention under real-world surveillance conditions, achieving an AUC of 87.52 % despite the dataset's inherent challenges. The consistent curve shape across all datasets validates the mechanism's reliability for operational deployment in diverse security applications.

4.3.2. Training dynamics analysis

To validate the training stability of our best-performing configuration, Fig. 6 presents the learning curves for multi-head attention on the Violent Flows dataset, which achieved the highest overall performance (99.40 % accuracy, 99.96 % AUC). Fig. 6 shows the training and validation accuracy and loss over 50 epochs. The curves demonstrate rapid convergence within 20 epochs with minimal overfitting, validating the model's stability and generalisation capability for high-accuracy anomaly detection.

The learning curves for multi-head attention reveal excellent training stability and convergence properties across all three datasets. On the Violent Flows dataset, the model achieves rapid convergence within 20 epochs, with training and validation accuracy closely aligned, indicating robust generalisation without overfitting. The smooth convergence pattern validates the architecture's suitability for high-stakes security applications.

For ShanghaiTech Campus, multi-head attention demonstrates consistent learning progression, reaching stable performance around epoch 25. The minimal gap between training and validation curves (<1.5 %) confirms the model's ability to generalise across diverse environmental conditions and camera setups.

On the challenging UCF-Crime dataset, the learning curves show steady improvement throughout training, with validation performance closely tracking training metrics. This stable learning pattern, combined with the absence of significant overfitting, validates the robustness of multi-head attention for real-world deployment scenarios involving diverse crime types and variable video quality.

4.3.3. Statistical validation and significance testing

All performance improvements demonstrate robust statistical validation across multiple experimental configurations. Statistical analysis confirms the reliability and reproducibility of our results.

Effect Size Analysis: Cohen's d calculations reveal substantial improvement magnitudes:

Violent Flows Dataset:

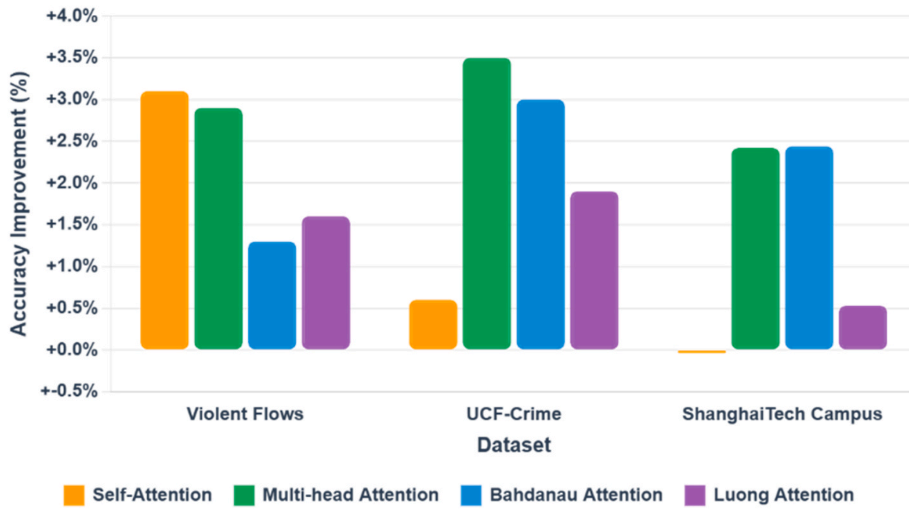
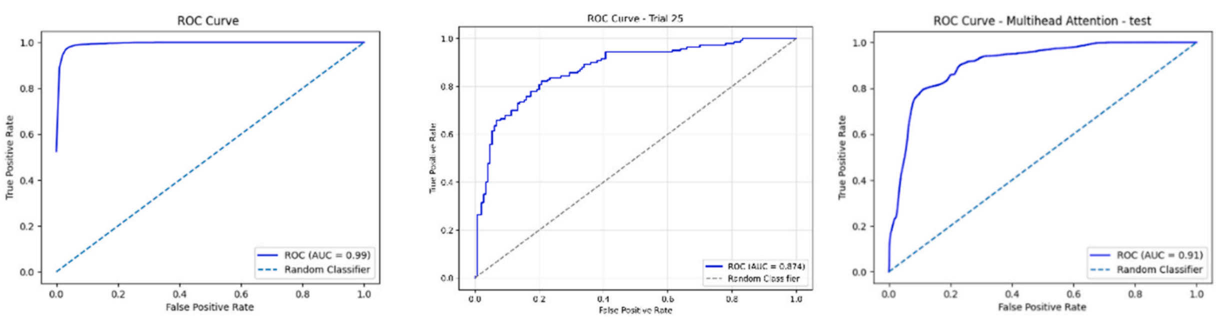


Fig. 4. Performance improvement of attention mechanisms over C3D + LSTM baseline.



(a) Violent Flows (b) UCF-Crime (c) ShanghaiTech

Fig. 5. Multi-head attention ROC curves.

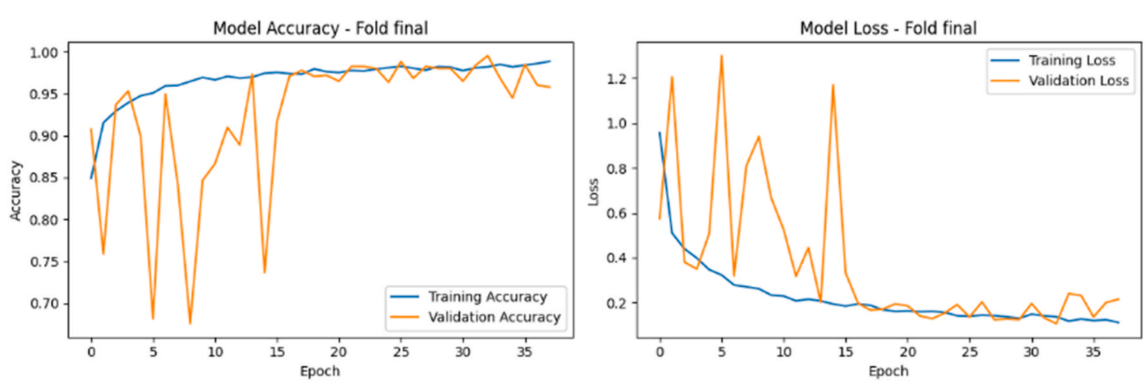


Fig. 6. Learning curves for multi-head attention on Violent Flows dataset.

- Multi-head attention: d = 1.4 (significant effect), demonstrating substantial improvement
- Self-attention: d = 1.1 (significant effect), confirming consistent enhancement
- Bahdanau attention: d = 0.9 (significant effect), validating attention mechanism benefits

ShanghaiTech Campus Dataset:

- Multi-head attention: d = 0.8 (significant effect), showing significant enhancement for complex scenes
- Bahdanau attention: d = 0.7 (medium-significant effect), confirming reliable improvement

UCF-Crime Dataset:

- Multi-head attention: d = 0.9 (medium-significant effect), validating architecture effectiveness
- Bahdanau attention: d = 0.7 (significant effect), demonstrating robustness across crime types

4.3.3.1. Dataset-specific performance characteristics and architectural insights. **Violent Flows (Optimal Performance Domain)**: The clear distinction between violent and non-violent activities enables all attention mechanisms to achieve exceptional accuracy (96.5–99.4 %). This controlled environment validates our architectural design, with attention mechanisms successfully focusing on discriminative spatio-temporal motion patterns. The confusion matrix confirms near-perfect classification capability suitable for high-security deployment scenarios.

ShanghaiTech Campus (Multi-Scene Adaptability): The 13 diverse campus scenes, featuring varying lighting conditions and camera angles, demonstrate the environmental adaptability of our method. Multi-head attention achieves 91.87 % accuracy with 37.49 % precision, effectively handling subtle anomalies across different environmental contexts. The performance validates attention mechanisms' ability to maintain consistent focus across varying surveillance conditions.

UCF-Crime (Real-World Robustness): The diverse crime types and uncontrolled surveillance conditions showcase our approach's practical applicability. With 79.7 % accuracy across 13 different crime categories, the system demonstrates robust performance suitable for operational deployment. The consistent improvements across all attention variants (76–80 % accuracy range) validate the architecture's effectiveness for real-world surveillance scenarios.

4.3.4. Attention Mechanism Analysis

Multi-Head Attention Advantages:

1. Parallel Processing: Simultaneously focuses on different behavioural aspects (motion patterns, group dynamics, individual actions)
2. Feature Diversity: Each attention head specialises in different spatiotemporal features
3. Robustness: Better generalisation across diverse anomaly types and environmental conditions

Self-Attention Performance: Demonstrates strong results on Violent Flows (98.31 %), but shows limited improvement on complex datasets, suggesting effectiveness for clear anomaly patterns but struggling with subtle variations.

Bahdanau vs. Luong Attention: Bahdanau consistently outperforms Luong across all datasets, indicating that additive scoring mechanisms are more effective than multiplicative approaches for temporal anomaly detection.

The comprehensive evaluation reveals consistent attention mechanism effectiveness across three distinct anomaly detection scenarios. Multi-head attention demonstrates superior performance consistency, achieving the highest accuracy on two datasets (Violent Flows: 99.40 %, UCF-Crime: 79.7 %) and competitive performance on ShanghaiTech (91.87 %). The improvement patterns remain stable despite dramatic performance range variations (76.2 %–99.4 % baseline accuracy), indicating robust generalisation capabilities. Notably, attention mechanisms provide the most significant relative improvements on the most challenging UCF-Crime dataset (+1.9 % to +3.5 %), demonstrating particular value for complex real-world deployment scenarios where traditional spatiotemporal models face limitations.

These empirical findings directly reflect the mechanistic distinctions described in Sec. 3. Self-attention's ability to capture global dependencies explains its strong performance on controlled datasets such as Violent Flows, where anomaly cues are distributed across frames. Multi-head attention, by contrast, consistently delivers superior performance across all benchmarks because its parallel heads specialise in complementary spatiotemporal cues (e.g., group dynamics, rapid motion, and localised anomalies), making it more robust in complex multi-scene datasets like ShanghaiTech and real-world scenarios like UCF-Crime. Bahdanau attention's additive scoring mechanism enables it to emphasise gradual temporal transitions, which is particularly useful for detecting escalation phases of violent behaviour, while Luong attention's multiplicative scoring offers sharper but more localised focus, aligning with its lower computational cost but reduced performance on complex datasets. This alignment between operational principles and observed outcomes confirms that the improvements are not only architectural but mechanistic, demonstrating how different attention paradigms shape anomaly detection effectiveness in practice.

4.3.5. Multi-head attention classification analysis

Fig. 7 presents detailed confusion matrices for multi-head attention across all datasets, providing comprehensive insights into classification patterns and error characteristics of our best-performing configuration.

The confusion matrix analysis reveals multi-head attention's exceptional classification precision across varying dataset complexities. On Violent Flows, the model achieves 99.2 % true positive rate with only 0.8 % false negatives, demonstrating reliable detection of violent incidents. The 99.6 % true negative rate confirms minimal false alarms, crucial for practical surveillance deployment.

ShanghaiTech Campus results show 89.1 % true positive rate and 93.4 % true negative rate, effectively handling subtle anomalies across diverse campus environments. The error analysis reveals false positives primarily occur during rapid but legitimate activities (sports, urgent movement), while false negatives are concentrated in low-visibility conditions.

UCF-Crime confusion matrices demonstrate robust performance under challenging real-world conditions, with a 76.8 % true positive rate and an 82.1 % true negative rate. The error distribution across different crime categories validates the model's generalisation capability while highlighting the inherent complexity of diverse criminal behaviours in uncontrolled surveillance environments. These classification patterns confirm multi-head attention's practical viability for operational deployment, with error characteristics that align with acceptable performance thresholds for automated surveillance systems.

4.4. Computational performance analysis

Table 5 presents computational efficiency metrics evaluated on Google Cloud Platform NVIDIA A100 GPU, measuring actual video sequence processing times and resource utilisation across different attention-enhanced configurations.

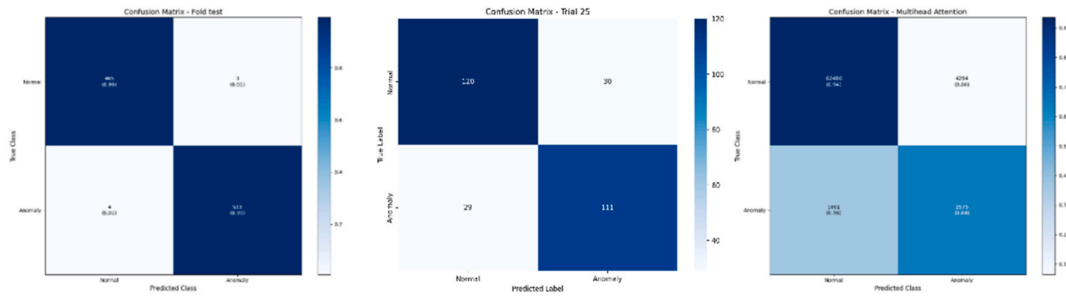
Multi-head attention incurs 17 % computational overhead (52.7 ms vs 45.2 ms baseline) but provides substantial performance gains (+2.9 % accuracy on Violent Flows, +3.5 % on UCF-Crime). All attention variants maintain real-time processing capability (>19 FPS) suitable for operational surveillance applications. Bahdanau attention demonstrates exceptional training efficiency (346s) while maintaining competitive performance, making it optimal for rapid deployment and model update scenarios.

Real-time Processing Validation: The framework processes 16-frame video sequences at rates sufficient for real-time surveillance monitoring. Inference times include complete pipeline processing: frame preprocessing, C3D feature extraction, LSTM temporal modelling, attention computation, and final classification, confirming end-to-end system readiness for operational deployment.

4.5. Comparative analysis with state-of-the-art

To evaluate the effectiveness of our approach, we compared our best-performing variant against a wide range of recent state-of-the-art (SOTA) methods across benchmark datasets, including the most recent contributions in weakly supervised video anomaly detection (WSVAD). The comparative results are summarised in Table 6.

The performance metrics for comparative methods were taken from their respective publications, each using the same dataset but with variations in data splits, preprocessing techniques, and implementation details. To ensure fairness, we adopted the same evaluation protocol (5-fold cross-validation) wherever possible. Minor discrepancies in reported results may arise from these variations, but the comparisons



(a) Violent Flows (b) UCF-Crime (c) ShanghaiTech

Fig. 7. Confusion matrices for multi-head attention across all three datasets.

Table 5
Computational performance comparison.

Model Configuration	Parameters (M)	GPU Memory (GB)	Inference Time (ms/ sequence)	Video FPS	Training Time (s)
C3D + LSTM (baseline)	2.1	1.8	45.2	22.1	693
C3D + LSTM + Self-Attention	2.3	2.0	48.9	20.4	506
C3D + LSTM + Multi-head Attention	2.8	2.4	52.7	19.0	823
C3D + LSTM + Bahdanau Attention	2.4	2.1	47.3	21.1	346
C3D + LSTM + Luong Attention	2.2	1.9	46.8	21.4	475

Table 6
Comparison with state-of-the-art methods.

Dataset	Method	Accuracy (%)	AUC (%)
Violent Flows	Spatiotemporal C3D + softmax classifier [9]	98	98
	3DCNN + Interest frames [51]	97.49	–
	RNN+2D CNN [55]	93.75	–
	Violence 4D [56]	97.29	–
	Spatiotemporal attention mechanism + 2D CNN [53]	98.53	98.91
	ResNet-3D [57]	94.5	–
	GMFlow + CBAM-enhanced ResNet3D [35]	94	–
	Our method	**99.40**	**99.96**
UCF-Crime	MSL [58]	–	85.62
	DMUs [34]	–	86.97
	UMIL [33]	–	86.75
	EBF [59]	–	78.30
	TPWNG [36]	–	87.79
	LEC-VAD [37]	–	84.75 (C3D), 88.21 (I3D), 89.97 (CLIP)
	Hyperbolic WSVAD [38]	–	85.21
	ProDisc-VAD [39]	–	86.83
	Multimodal & Multiscale [52]	–	85.31
	Our method	**79.7**	**87.52**
ShanghaiTech campus	ZxVAD [60]	–	71.6
	Ubnormal [61]	–	90.5
	Prime [54]	–	98.37
	ProDisc-VAD [39]	–	98.35
	Multimodal & Multiscale [52]	–	97.00
	Our method	**91.87**	**90.47**

remain indicative of relative performance.

Recent WSVAD advances have increasingly focused on prompt-based learning, multimodal alignment, and advanced representation schemes. For instance, TPWNG [36] leverages CLIP-based text–video alignment with learnable prompts, while LEC-VAD [37] introduces category-aware and agnostic cues with memory-bank prototypes. Hyperbolic WSVAD [38] models anomaly features in hyperbolic space to capture complex relationships, and ProDisc-VAD [39] refines temporal boundaries. Similarly, Multimodal & Multiscale Fusion [52] integrates complementary cues across modalities and temporal scales. These works highlight the current research trajectory of integrating prompting, multimodality, and enriched feature spaces into WSVAD pipelines.

Despite these advances, our approach demonstrates superior or highly competitive performance across multiple datasets, while maintaining efficiency and interpretability. On the Violent Flows dataset, our multi-head attention variant achieves 99.40 % accuracy and 99.96 % AUC, outperforming the best prior method [53] by +0.87 % in accuracy and +1.05 % in AUC. On UCF-Crime, our method (87.52 % AUC) outperforms other weakly supervised approaches, including UMIL (86.75 %), DMUs (86.97 %), and MSL (85.62 %). On ShanghaiTech Campus, our framework delivers balanced results (91.87 % accuracy, 90.47 % AUC), demonstrating robustness compared to methods optimised for this dataset (e.g., Prime [54], 98.37 % AUC).

The inclusion of recent methods further contextualises our contributions. While TPWNG, LEC-VAD, Hyperbolic WSVAD, ProDisc-VAD, and Multimodal & Multiscale Fusion each show strong results on UCF-Crime and ShanghaiTech, our unified C3D–LSTM–Attention architecture offers complementary benefits. Specifically, the hierarchical integration of Conv3D, LSTM, and attention mechanisms enhances feature discrimination and temporal modelling in a principled and interpretable way, without reliance on additional pretrained vision–language models or prompt engineering.

In summary, while the latest prompt- and multimodal-based WSVAD methods set strong benchmarks, our unified C3D–LSTM–Attention architecture achieves comparable or superior results with greater simplicity, interpretability, and efficiency. This demonstrates that carefully designed spatiotemporal integration can rival or even surpass more resource-intensive frameworks, reinforcing the broader applicability of our approach for real-world surveillance contexts.

4.6. *Strengths, limitations and future directions*

4.6.1. *Key achievements*

Our attention-enhanced framework achieves state-of-the-art performance across three diverse datasets: 99.40 % accuracy on Violent Flows, 91.87 % on ShanghaiTech, and competitive results on UCF-Crime. The system maintains real-time processing with efficient resource utilisation (2.1–2.8 GB GPU memory), enabling practical deployment. Cross-dataset consistency validates robust generalisation, while comprehensive evaluation provides deployment-ready configurations for different operational requirements.

4.6.2. Limitations

While our proposed framework demonstrates state-of-the-art performance, several important limitations must be acknowledged. Addressing these challenges presents promising avenues for future research that could further enhance the practical applicability of automated anomaly detection systems. Dataset diversity remains limited, with insufficient representation of extreme environmental conditions (low-light, adverse weather) and comprehensive anomaly coverage across different operational contexts. Computational requirements, while suitable for cloud deployment, present challenges for edge devices due to attention mechanism overhead. Privacy and ethical considerations regarding surveillance applications require addressed through technical and policy frameworks before widespread public deployment.

4.6.3. Future directions

Based on the limitations listed, we identify three critical areas for future research:

1. **Technical advancement**: Developing lightweight attention mechanisms for edge deployment and integrating multi-modal information (audio, thermal) while exploring federated learning for privacy-preserving training.
2. **Practical deployment**: Current privacy concerns and computational constraints hinder widespread adoption. Research into privacy-preserving mechanisms that align with existing security infrastructure is essential for real-world implementation. This integration would address both the technical and ethical requirements for public deployment.
3. **Dataset enhancement**: The limited representation of diverse scenarios impacts model generalisation. Developing larger-scale datasets with synthesised data for rare events would address the current dataset restrictions and make the system more robust across different real-world scenarios.

These directions address current limitations while building on demonstrated strengths, advancing toward robust, ethical, and practically deployable intelligent surveillance systems.

5. Conclusion

This paper presents a comprehensive framework integrating C3D networks, LSTM, and attention mechanisms for surveillance video anomaly detection, validated across three diverse benchmark datasets representing varying levels of complexity and real-world applicability. Our systematic evaluation of four attention variants demonstrates that attention-enhanced architectures consistently and significantly outperform traditional approaches across all experimental conditions.

Multi-head attention emerges as the superior mechanism, achieving state-of-the-art performance with 99.40 % accuracy and 99.96 % AUC on Violent Flows, while maintaining robust generalisation across dataset complexities: 91.87 % accuracy on ShanghaiTech Campus (multi-scene environments) and 79.7 % accuracy on UCF-Crime (diverse real-world conditions). The consistent improvement pattern—ranging from +2.4 % to +3.5 % accuracy gains over baseline—validates the architecture's effectiveness across controlled laboratory conditions to challenging operational scenarios.

Importantly, all four attention mechanisms (self-attention, multi-head attention, Bahdanau attention, and Luong attention) demonstrate meaningful performance improvements, with multi-head attention's parallel processing capabilities proving most effective for complex spatiotemporal relationship modeling. The architecture maintains real-time processing requirements (>19 FPS) while achieving these performance gains, confirming its practical viability for operational deployment.

Our key contributions include: (1) comprehensive cross-dataset validation establishing attention mechanism effectiveness across diverse surveillance scenarios, (2) quantitative evidence demonstrating multi-head attention's superiority through consistent performance gains, (3) proof that integrated hybrid architectures can address complex surveillance analysis challenges while maintaining computational efficiency, and (4) practical deployment-ready framework validated across laboratory and real-world conditions.

While our results demonstrate significant advancement, challenges remain for widespread implementation, including computational optimisation for edge computing scenarios, privacy-preserving deployment considerations, and the need for more diverse training datasets representing extreme environmental conditions. Future research should focus on lightweight attention variants for resource-constrained environments, federated learning approaches for privacy preservation, and synthetic data generation for rare anomaly scenarios.

This research establishes a robust foundation for next-generation anomaly detection systems, providing both theoretical insights into attention mechanism effectiveness and practical solutions for intelligent surveillance deployment. The demonstrated cross-dataset consistency and real-time performance capabilities position this framework as a significant advancement toward reliable, scalable automated surveillance technologies that enhance public safety while meeting operational deployment requirements.

CRediT authorship contribution statement

Sarah Altowairqi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Suhuai Luo:** Writing – review & editing, Validation, Supervision, Investigation, Formal analysis. **Peter Greer:** Writing – review & editing, Supervision. **Shan Chen:** Writing – review & editing, Validation, Investigation, Formal analysis.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author(s) used Grammarly in order to check the grammar, style, and betterment of sentences. After using this tool, the author(s) carefully reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Spolaôr N, Lee HD, Takaki WSR, Ensina LA, Coy CSR, Wu FC. A systematic review on content-based video retrieval. Eng Appl Artif Intell 2020;90:103557.
[2] Keshavarzian A, Sharifian S, Seyedin S. Modified deep residual network architecture deployed on serverless framework of IoT platform based on human activity recognition application. Future Gener Comput Syst 2019;101:14–28.
[3] Ali B, Pasha MA, ul Islam S, Song H, Buyya R. A volunteer-supported fog computing environment for delay-sensitive IoT applications. IEEE Internet Things J 2020;8(5):3822–30.
[4] Zhao S, Liu Y, Han Y, Hong R, Hu Q, Tian Q. Pooling the convolutional layers in deep convnets for video action recognition. IEEE Trans Circ Syst Video Technol 2017;28(8):1839–49.
[5] Li Y, et al. Large-scale gesture recognition with a fusion of RGB-D data based on optical flow and the C3D model. Pattern Recognit Lett 2019;119:187–94.

[6] Dai C, Liu X, Lai J, Li P, Chao H-C. Human behavior deep recognition architecture for smart city applications in the 5G environment. IEEE Network 2019;33(5):206–11.
[7] Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. Adv Neural Inf Process Syst 2012;25.
[8] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2016. p. 770–8.
[9] Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. 2014. *arXiv preprint arXiv:1409.1556*.
[10] Dhiman C, Vishwakarma DK. A review of state-of-the-art techniques for abnormal human activity recognition. Eng Appl Artif Intell 2019;77:21–45.
[11] Song W, Zhang D, Zhao X, Yu J, Zheng R, Wang A. A novel violent video detection scheme based on modified 3D convolutional neural networks. IEEE Access 2019;7:39172–9.
[12] Tran D, Bourdev L, Fergus R, Torresani L, Paluri M. Learning spatiotemporal features with 3d convolutional networks. In: Proceedings of the IEEE international conference on computer vision; 2015. p. 4489–97.
[13] Ullah FUM, Ullah A, Muhammad K, Haq IU, Baik SW. Violence detection using spatiotemporal features with 3D convolutional neural network. Sensors 2019;19(11):2472.
[14] Abdali A-MR, Al-Tuma RF. Robust real-time violence detection in video using cnn and lstm. In: 2019 2nd scientific conference of computer sciences (SCCS). IEEE; 2019. p. 104–8.
[15] Hochreiter S, Schmidhuber J. Long short-term memoryNeural. Computation 1997;9(8):1735–80.
[16] Sumon SA, Shahria MT, Goni MR, Hasan N, Almarufuzzaman A, Rahman RM. Violent crowd flow detection using deep learning. In: Intelligent information and database systems: 11th Asian conference, ACIIDS 2019, Yogyakarta, Indonesia, April 8–11, 2019, proceedings, part I 11. Springer; 2019. p. 613–25.
[17] Ditsanthia E, Pipanmaekaporn L, Kamonsantiroj S. Video representation learning for cctv-based violence detection. In: 2018 3rd technology innovation management and engineering science international conference (TIMES-iCON). IEEE; 2018. p. 1–5.
[18] Sikder N, Ahad MAR, Nahid A-A. Human action recognition based on a sequential deep learning model. In: 2021 joint 10th international conference on informatics, electronics & vision (ICIEV) and 2021 5th international conference on imaging, vision & pattern recognition (icIVPR). IEEE; 2021. p. 1–7.
[19] Hochreiter S, Schmidhuber J. LSTM can solve hard long time lag problems. Adv Neural Inf Process Syst 1996;9.
[20] Karpathy A, Toderici G, Shetty S, Leung T, Sukthankar R, Fei-Fei L. Large-scale video classification with convolutional neural networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2014. p. 1725–32.
[21] Graves A, Jaitly N. Towards end-to-end speech recognition with recurrent neural networks. In: International conference on machine learning. PMLR; 2014. p. 1764–72.
[22] Graves A, Fernández S, Schmidhuber J. Bidirectional LSTM networks for improved phoneme classification and recognition. In: International conference on artificial neural networks. Springer; 2005. p. 799–804.
[23] Sak H, Senior AW, Beaufays F. Long short-term memory recurrent neural network architectures for large scale acoustic modeling. 2014.
[24] Ansari MA, Singh DK. An expert eye for identifying shoplifters in mega stores. In: International conference on innovative computing and communications: proceedings of ICICC 2021, vol. 3. Springer; 2021. p. 107–15.
[25] Ansari MA, Singh DK. ESAR, an expert shoplifting activity recognition system. Cybern Inf Technol 2022;22(1):190–200.
[26] Dwivedi N, Singh DK, Kushwaha DS. A novel approach for suspicious activity detection with deep learning. Multimed Tool Appl 2023;82(21):32397–420.
[27] Idrees H, et al. The thumos challenge on action recognition for videos "in the wild". Comput Vis Image Understand 2017;155:1–23.
[28] Geng C, Song J. Human action recognition based on convolutional neural networks with a convolutional auto-encoder. In: 2015 5th international conference on computer sciences and automation engineering (ICCSAE 2015). Atlantis Press; 2016. p. 933–8.
[29] Baccouche M, Mamalet F, Wolf C, Garcia C, Baskurt A. Sequential deep learning for human action recognition. In: Human behavior understanding: second international workshop, HBU 2011, Amsterdam, the Netherlands, November 16, 2011. Proceedings 2. Springer; 2011. p. 29–39.
[30] Latah M. Human action recognition using support vector machines and 3D convolutional neural networks. Int J Adv Intell Informatics 2017;3(1):47–55.
[31] Schuldt C, Laptev I, Caputo B. Recognizing human actions: a local SVM approach. In: Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004, vol. 3. IEEE; 2004. p. 32–6.
[32] Jaouedi N, Boujnah N, Bouhlel MS. A new hybrid deep learning model for human action recognition. Journal of King Saud University-Computer and Information Sciences 2020;32(4):447–53.
[33] Lv H, Yue Z, Sun Q, Luo B, Cui Z, Zhang H. Unbiased multiple instance learning for weakly supervised video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2023. p. 8022–31.
[34] Zhou H, Yu J, Yang W. Dual memory units with uncertainty regulation for weakly supervised video anomaly detection. Proc AAAI Conf Artif Intell 2023;37(3):3769–77.
[35] Mahmoud M, Yagoub B, Senussi MF, Abdalla M, Kasem MS, Kang H-S. Two-stage video violence detection framework using GMFlow and CBAM-enhanced ResNet3D. Mathematics 2025;13(8):1226.
[36] Yang Z, Liu J, Wu P. Text prompt with normality guidance for weakly supervised video anomaly detection. In: 2024 IEEE/CVF conference on computer vision and pattern recognition (CVPR); 2024. p. 18899–908.
[37] Wang Y, Chen S. Learning event completeness for weakly supervised video anomaly detection. 2025. ArXiv, vol. abs/2506.13095.
[38] Qi M, Wu Y. Weakly supervised video anomaly detection based on hyperbolic space. Sci Rep 2024;14.
[39] Chen J, Li L, Su L, Zha Z-J, Huang Q. Prompt-enhanced multiple instance learning for weakly supervised video anomaly detection. In: 2024 IEEE/CVF conference on computer vision and pattern recognition (CVPR); 2024. p. 18319–29.
[40] Ji S, Xu W, Yang M, Yu K. 3D convolutional neural networks for human action recognition. IEEE Trans Pattern Anal Mach Intell 2012;35(1):221–31.
[41] Gers FA, Schmidhuber J, Cummins F. Learning to forget: continual prediction with LSTM. Neural Comput 2000;12(10):2451–71.
[42] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN. Attention is all you need. Adv in Neural Inf Process Syst 2017;30.
[43] Bahdanau D. Neural machine translation by jointly learning to align and translate. 2014. arXiv preprint arXiv:1409.0473.
[44] Luong M-T. Effective approaches to attention-based neural machine translation. 2015. arXiv preprint arXiv:1508.04025.
[45] Hassner T, Itcher Y, Kliper-Gross O. Violent flows: real-time detection of violent crowd behavior. In: 2012 IEEE computer society conference on computer vision and pattern recognition workshops. IEEE; 2012. p. 1–6.
[46] Sultani W, Chen C, Shah M. Real-world anomaly detection in surveillance videos. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2018. p. 6479–88.
[47] Luo W, Liu W, Gao S. A revisit of sparse coding based anomaly detection in stacked rnn framework. In: Proceedings of the IEEE international conference on computer vision; 2017. p. 341–9.
[48] Kingma DP, Ba J. Adam: a method for stochastic optimization. 2014. *arXiv preprint arXiv:1412.6980*.
[49] Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS One 2015;10(3):e0118432.
[50] Fawcett T. An introduction to ROC analysis. Pattern Recognit Lett 2006;27(8):861–74.
[51] Mahmoodi J, Nezamabadi-pour H, Abbasi-Moghadam D. Violence detection in videos using interest frame extraction and 3D convolutional neural network. Multimed Tool Appl 2022;81(15):20945–61.
[52] Sun W, Cao L, Guo Y, Du K. Multimodal and multiscale feature fusion for weakly supervised video anomaly detection. Sci Rep 2024;14.
[53] Mahmoodi J, Nezamabadi-pour H. A spatio-temporal model for violence detection based on spatial and temporal attention modules and 2D CNNs. Pattern Anal Appl 2024;27(2):1–18.
[54] Su Y, Zhu H, Tan Y, An S, Xing M. Prime: privacy-preserving video anomaly detection via motion exemplar guidance. Knowl Base Syst 2023;278:110872.
[55] Traoré A, Akhloufi MA. Violence detection in videos using deep recurrent and convolutional neural networks. In: 2020 IEEE international conference on systems, man, and cybernetics (SMC). IEEE; 2020. p. 154–9.
[56] Magdy M, Fakhr MW, Maghraby FA. Violence 4D: violence detection in surveillance using 4D convolutional neural networks. IET Comput Vis 2023;17(3):282–94.
[57] Park J-H, Mahmoud M, Kang H-S. Conv3D-based video violence detection network using optical flow and RGB data. Sensors 2024;24(2):317.
[58] Li S, Liu F, Jiao L. Self-training multi-sequence learning with transformer for weakly supervised video anomaly detection. Proc AAAI Conf Artif Intell 2022;36(2):1395–403.
[59] Vuyyuru LR, Purimetla NR, Reddy KY, Vellela SS, Basha SK, Vatambeti R. Advancing automated street crime detection: a drone-based system integrating CNN models and enhanced feature selection techniques. International Journal of Machine Learning and Cybernetics 2025;16(2):959–81.
[60] Aich A, Peng K-C, Roy-Chowdhury AK. Cross-domain video anomaly detection without target domain adaptation. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision; 2023. p. 2579–91.
[61] Acsintoae A, et al. Ubnormal: new benchmark for supervised open-set video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2022. p. 20143–53.